

ZERUI GUO

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EDUCATION

- University of Wisconsin-Madison, USA** *Jun. 2023 - May 2027 (Expected)*
Ph.D. student in Computer Science, Advisor: Ming Liu
- Beihang University (BUAA), China** *Sep. 2020 - Jan. 2023*
Master of Computer Science and Technology, Advisor: Yuebin Bai
- Beijing University of Posts and Telecommunications (BUPT), China** *Sep. 2016 - Jun. 2020*
Bachelor of Network Engineering, Ranking: 4/135
School Outstanding Graduate

RESEARCH INTERESTS

Hardware-Aware Networked Systems; Performance Engineering; Disaggregated Computing

PUBLICATIONS

- Building A CSFQ-Inspired Transport for Switched CXL Memory Pooling** *NSDI 2026*
Zerui Guo, Emily Shriver, and Ming Liu
- Understanding and Profiling CXL.mem Using PathFinder** *SIGCOMM 2025*
Xiao Li, Zerui Guo, Yuebin Bai, Mahesh Ketkar, Hugh Wilkinson, and Ming Liu
- Building Massive MIMO Baseband Processing on a Single-Node Supercomputer** *NSDI 2025*
Xincheng Xie, Wentao Hou, Zerui Guo, and Ming Liu
- LogNIC: A High-Level Performance Model for SmartNICs** *MICRO 2023*
Zerui Guo, Jiaxin Lin, Yuebin Bai, Daehyeok Kim, Michael Swift, Aditya Akella, and Ming Liu
- LEED: A Low-Power, Fast Persistent Key-Value Store on SmartNIC JBOFs** *SIGCOMM 2023*
Zerui Guo, Hua Zhang, Chenxingyu Zhao, Yuebin Bai, Michael Swift, and Ming Liu
- A Millimeter-wave Multi-channel MAC with Dynamic Spectrum Access Capability for Mobile Self-organizing Heterogeneous Networks** *IPCCC 2021*
Peng Feng, Yuebin Bai, Zerui Guo, Xiaolin Wang, Yining Zhang, Bojian Bai, and Junmin Zhang

RESEARCH EXPERIENCE

- MemChannel: Transport for Switched CXL Memory Pooling** *Jun. 2023 - Apr. 2026*
University of Wisconsin-Madison
- Designed and implemented a CSFQ-inspired transport for switched CXL memory pooling, managing bandwidth among competing memory flows.
 - Developed host-side rate control and fabric coordination mechanisms to mitigate intra-host contention and cross-host interference.
 - Demonstrated performance isolation and $2\text{--}3.5\times$ application-performance improvements in multi-tenant evaluations on a real switched CXL memory pool.

LogNIC: A High-Level Performance Model for SmartNICs

Feb. 2022 - Apr. 2023

University of Wisconsin-Madison

- Developed a packet-centric model that represents SmartNIC-offloaded programs as execution graphs and estimates latency and throughput across traffic profiles.
- Evaluated commodity SmartNICs and an academic prototype across five application scenarios to estimate performance bounds and assess software and hardware design choices.
- Model-guided parallelism tuning increased throughput by up to 36.4% and reduced latency by up to 22.8% versus heuristic approaches across five evaluated microservice workloads.

LEED: A Low-Power, Fast Persistent Key-Value Store on SmartNIC JBOFs

Sep. 2021 - Feb. 2023

University of Wisconsin-Madison

- Designed a distributed, replicated persistent key-value store for SmartNIC JBOFs that trades abundant I/O bandwidth for constrained compute and memory resources.
- Developed early I/O scheduling and storage optimizations across the per-SSD, intra-JBOF, and inter-JBOF layers.
- Achieved up to $4.2\times$ the requests per Joule of Kvell on server JBOFs and up to $19.1\times$ that of FAWN on embedded storage nodes in evaluated 256B/1KB workloads.

TEACHING EXPERIENCE

Teaching Assistant, Embedded Systems, BUAA

2020 Fall, 2021 Spring

Led firmware-programming and Linux device-driver labs.

Teaching Assistant, Wireless Network Systems, BUAA

2020 Fall

Supported machine-learning-based extensions to AODV-uu routing.

Teaching Assistant, Introduction of Intelligent Car, BUPT

2019 Fall

Taught approximately 30 students and assessed line-tracking car projects.

WORK EXPERIENCE

University of Wisconsin-Madison, USA

Jun. 2023 - Present

Research Assistant

HONORS & AWARDS

- Beihang University Scholarship, Second Prize 2021
- Interdisciplinary Contest in Modeling, Honorable Mention; China “Internet Plus” Contest, Second Prize 2019
- NXP Cup Intelligent Car Race, Third Prize 2018
- BUPT Scholarship, Second Prize 2016–2018

SKILLS

Programming:	C/C++, Python, Rust, Golang, JavaScript, Julia
Frameworks & Tools:	SPDK, RDMA, UCX, NVENC, Linux Networking, KVM, Docker
Research Software (open source):	MemChannel, LEED, LogNIC, MegaStation, PathFinder